

This guideline replaces CG68.

This guideline is the basis of QS2.

Overview

This guideline covers interventions in the acute stage of a stroke or transient ischaemic attack (TIA). It offers the best clinical advice on the diagnosis and acute management of stroke and TIA in the 48 hours after onset of symptoms.

Who is it for?

- Healthcare professionals in primary and secondary NHS healthcare settings
- Commissioners and providers of services
- People aged over 16 who have had a stroke or TIA, their families and carers

Recommendations

People have the right to be involved in discussions and make informed decisions about their care, as described in [NICE's information on making decisions about your care](#).

[Making decisions using NICE guidelines](#) explains how we use words to show the strength (or certainty) of our recommendations, and has information about prescribing medicines (including off-label use), professional guidelines, standards and laws (including on consent and mental capacity), and safeguarding.

[NICE has also produced patient decision aids on decompressive hemicraniectomy](#).

1.1 Rapid recognition of symptoms and diagnosis

Prompt recognition of symptoms of stroke and transient ischaemic attack

- 1.1.1 Use a validated tool, such as FAST (Face Arm Speech Test), outside hospital to screen people with sudden onset of neurological symptoms for a diagnosis of stroke or transient ischaemic attack (TIA). **[2008]**
- 1.1.2 Exclude hypoglycaemia in people with sudden onset of neurological symptoms as the cause of these symptoms. **[2008]**
- 1.1.3 For people who are admitted to the emergency department with a suspected stroke or TIA, establish the diagnosis rapidly using a validated tool, such as ROSIER (Recognition of Stroke in the Emergency Room). **[2008]**

Initial management of suspected and confirmed TIA

- 1.1.4 Offer aspirin (300 mg daily), unless contraindicated, to people who have had a

suspected TIA, to be started immediately. **[2019]**

For a short explanation of why the committee made this 2019 recommendation and how it might affect practice, see the [rationale and impact section on initial management of suspected and confirmed transient ischaemic attack \(aspirin\)](#).

Full details of the evidence and the committee's discussion are in [evidence review A: aspirin](#).

- 1.1.5 Refer immediately people who have had a suspected TIA for specialist assessment and investigation, to be seen within 24 hours of onset of symptoms. **[2019]**
- 1.1.6 Do not use scoring systems, such as ABCD2, to assess risk of subsequent stroke or to inform urgency of referral for people who have had a suspected or confirmed TIA. **[2019]**
- 1.1.7 Offer secondary prevention, in addition to aspirin, as soon as possible after the diagnosis of TIA is confirmed. **[2008, amended 2019]**

For a short explanation of why the committee made these 2019 recommendations and how they might affect practice, see the [rationale and impact section on initial management of suspected and confirmed transient ischaemic attack](#).

Full details of the evidence and the committee's discussion are in [evidence review B: TIA prediction rules](#).

1.2 Imaging for people who have had a suspected TIA or acute non-disabling stroke

Suspected TIA

- 1.2.1 Do not offer CT brain scanning to people with a suspected TIA unless there is

clinical suspicion of an alternative diagnosis that CT could detect. **[2019]**

- 1.2.2 After specialist assessment in the TIA clinic, consider MRI (including diffusion-weighted and blood-sensitive sequences) to determine the territory of ischaemia, or to detect haemorrhage or alternative pathologies. If MRI is done, perform it on the same day as the assessment. **[2019]**

For a short explanation of why the committee made these recommendations and how they might affect practice, see the [rationale and impact section on imaging for people who have had a suspected TIA or acute non-disabling stroke](#).

Full details of the evidence and the committee's discussion are in [evidence review C: TIA imaging](#).

Carotid imaging

- 1.2.3 Everyone with TIA who after specialist assessment is considered as a candidate for carotid endarterectomy should have urgent carotid imaging. **[2008, amended 2019]**

Urgent carotid endarterectomy

- 1.2.4 Ensure that people with stable neurological symptoms from acute non-disabling stroke or TIA who have symptomatic carotid stenosis of 50% to 99% according to the NASCET (North American Symptomatic Carotid Endarterectomy Trial) criteria, or 70% to 99% according to the ECST (European Carotid Surgery Trial) criteria:
- are assessed and referred urgently for carotid endarterectomy to a service following current national standards (see the [NHS England and NHS Improvement National Stroke Service Model](#))
 - receive best medical treatment (control of blood pressure, antiplatelet agents, cholesterol lowering through diet and drugs, lifestyle advice). **[2008, amended 2019]**

- 1.2.5 Ensure that people with stable neurological symptoms from acute non-disabling stroke or TIA who have symptomatic carotid stenosis of less than 50% according to the NASCET criteria, or less than 70% according to the ECST criteria:
- do not have surgery
 - receive best medical treatment (control of blood pressure, antiplatelet agents, cholesterol lowering through diet and drugs, lifestyle advice). **[2008]**
- 1.2.6 Ensure that carotid imaging reports clearly state which criteria (ECST or NASCET) were used when measuring the extent of carotid stenosis. **[2008]**

1.3 Specialist care for people with acute stroke

Specialist stroke units

- 1.3.1 Admit everyone with suspected stroke directly to a specialist acute stroke unit after initial assessment, from either the community, the emergency department, or outpatient clinics. (An acute stroke unit is a discrete area in the hospital that is staffed by a specialist stroke multidisciplinary team. It has access to equipment for monitoring and rehabilitating patients. Regular multidisciplinary team meetings occur for goal setting.). **[2008]**

Brain imaging for the early assessment of people with suspected acute stroke

- 1.3.2 Perform brain imaging immediately with a non-enhanced CT for people with suspected acute stroke if any of the following apply (see [additional information](#)):
- indications for thrombolysis or thrombectomy
 - on anticoagulant treatment
 - a known bleeding tendency
 - a depressed level of consciousness (Glasgow Coma Score below 13)

is considered. **[2008]**

- 1.4.18 For people with prosthetic valves who have disabling cerebral infarction and who are at significant risk of haemorrhagic transformation, stop anticoagulation treatment for 1 week and substitute aspirin 300 mg. **[2008]**
- 1.4.19 Ensure that people with ischaemic stroke and symptomatic proximal deep vein thrombosis or pulmonary embolism receive anticoagulation treatment in preference to treatment with aspirin unless there are other contraindications to anticoagulation. **[2008]**
- 1.4.20 Treat people who have haemorrhagic stroke and symptomatic deep vein thrombosis or pulmonary embolism to prevent the development of further pulmonary emboli using either anticoagulation or a caval filter. **[2008]**

Statin treatment

- 1.4.21 Immediate initiation of statin treatment is not recommended in people with acute stroke (see [additional information](#)). **[2008]**
- 1.4.22 Continue statin treatment in people with acute stroke who are already receiving statins. **[2008]**

1.5 Maintenance or restoration of homeostasis

Supplemental oxygen therapy

- 1.5.1 Give supplemental oxygen to people who have had a stroke only if their oxygen saturation drops below 95%. The routine use of supplemental oxygen is not recommended in people with acute stroke who are not hypoxic. **[2008]**

Be aware that some pulse oximeters can underestimate or overestimate oxygen saturation levels, especially if the saturation level is borderline. Overestimation has been reported in people with dark skin. See also the [NHS England Patient](#)

Safety Alert on the risk of harm from inappropriate placement of pulse oximeter probes.

Blood sugar control

- 1.5.2 Maintain a blood glucose concentration between 4 and 11 mmol/litre in people with acute stroke. **[2008]**
- 1.5.3 Provide optimal insulin therapy, which can be achieved by the use of intravenous insulin and glucose, to all adults with type 1 diabetes with threatened or actual stroke. Critical care and emergency departments should have a protocol for such management. **[2008]**

Blood pressure control for people with acute intracerebral haemorrhage

- 1.5.4 Consider rapid blood pressure lowering for people with acute intracerebral haemorrhage who do not have any of the exclusions listed in recommendation 1.5.7 and who:
- present within 6 hours of symptom onset **and**
 - have a systolic blood pressure of between 150 and 220 mmHg. **[2022]**
- 1.5.5 Taking into account the risk of harm, consider rapid blood pressure lowering on a case-by-case basis for people with acute intracerebral haemorrhage who do not have any of the exclusions listed in recommendation 1.5.7 and who:
- present beyond 6 hours of symptom onset **or**
 - have a systolic blood pressure greater than 220 mmHg. **[2022]**
- 1.5.6 When rapidly lowering blood pressure in people with acute intracerebral haemorrhage, aim to reach a systolic blood pressure of 140 mmHg or lower while ensuring that the magnitude drop does not exceed 60 mmHg within 1 hour of starting treatment. **[2022]**

1.5.7 Do not offer rapid blood pressure lowering to people who:

- have an underlying structural cause (for example, tumour, arteriovenous malformation or aneurysm)
- have a score on the Glasgow Coma Scale of below 6
- are going to have early neurosurgery to evacuate the haematoma
- have a massive haematoma with a poor expected prognosis. **[2019]**

1.5.8 When considering blood pressure lowering in young people aged 16 or 17 with acute intracerebral haemorrhage who do not have any of the exclusions listed in recommendation 1.5.7, seek advice from a paediatric specialist. **[2022]**

For a short explanation of why the committee made the 2022 recommendations and how they might affect practice, see the [rationale and impact section on blood pressure control for people with acute intracerebral haemorrhage](#).

Full details of the evidence and the committee's discussion are in [evidence review E: intensive interventions to lower blood pressure in people with acute intracerebral haemorrhage](#).

Blood pressure control for people with acute ischaemic stroke

1.5.9 Anti-hypertensive treatment in people with acute ischaemic stroke is recommended only if there is a hypertensive emergency with one or more of the following serious concomitant medical issues:

- hypertensive encephalopathy
- hypertensive nephropathy
- hypertensive cardiac failure/myocardial infarction
- aortic dissection
- pre-eclampsia/eclampsia. **[2008, amended 2019]**

- 1.5.10 Blood pressure reduction to 185/110 mmHg or lower should be considered in people who are candidates for intravenous thrombolysis. **[2008]**

1.6 Nutrition and hydration

Assessment of swallowing function

- 1.6.1 On admission, ensure that people with acute stroke have their swallowing screened by an appropriately trained healthcare professional before being given any oral food, fluid or medication. **[2008]**
- 1.6.2 If the admission screen indicates problems with swallowing, ensure that the person has a specialist assessment of swallowing, preferably within 24 hours of admission and not more than 72 hours afterwards. **[2008]**
- 1.6.3 People with suspected aspiration on specialist assessment, or who require tube feeding or dietary modification for 3 days, should be:
- re-assessed and considered for instrumental examination
 - referred for dietary advice. **[2008]**
- 1.6.4 People with acute stroke who are unable to take adequate nutrition, fluids and medication orally should:
- receive tube feeding with a nasogastric tube within 24 hours of admission unless they have had thrombolysis
 - be considered for a nasal bridge tube or gastrostomy if they are unable to tolerate a nasogastric tube
 - be referred to an appropriately trained healthcare professional for detailed nutritional assessment, individualised advice and monitoring
 - have their oral medication reviewed to amend either the formulation or the route of administration. **[2008, amended 2019]**

Terms used in this guideline

This section defines terms that have been used in a particular way for this guideline. For other definitions, see the [NICE glossary](#).

High-intensity mobilisation

High-intensity mobilisation refers to the very early mobilisation intervention from the [AVERT trial](#). (Further details of the intervention performed in the trial can be found in [NICE's evidence review F: very early mobilisation](#).) It includes mobilisation that:

- begins within the first 24 hours of stroke onset
- includes at least 3 additional out-of-bed sessions compared with usual care
- focuses on sitting, standing and walking (that is, out of bed) activity.